ActInf MathStream 009.1 ~ Jonathan Gorard: A computational perspective

MathStream | Jonathan Gorard: A computational perspective on observation, cognition | 1:25:24

Hello and welcome everyone. This is Active Inference Math Stream 9.1 on March 5th, 2024.

We're here with Jonathan Gorard and we'll be discussing a variety of topics yet to be determined,

or are they? So thank you for joining and to you for any introduction and we'll really look

forward to everyone's comments and questions. So thanks again for joining, to you.

Okay, well yeah, thanks very much, Daniel, for the introduction and for inviting me to be here on Active Inference.

I'm looking forward to a very, very fun discussion. So I don't have anything especially prepared to talk about,

which is probably a good thing because it means we'll be able to extend the unstructured part of this for as long as possible.

But I think just to give a little bit of context, I want to talk about an area where I think some things that I've been working on,

some collaborators that might have been working on, that might have some kind of intersection of interest with things that,

you know, active inference type people might care about, right?

So, and in particular, that concerns the relationship between kind of computation, observation and cognition,

and specifically using methods that come from category theory and topos theory and some other kind of branches of mathematics and theoretical computer science,

to understand the relationship between system, specifically the computational and algorithmic complexity of systems,

versus the computational algorithmic complexity of observers of those systems,

and how those things trade off between each other.

So, so just to give a little bit of context to that, I want to show, these are just some visuals from a paper that I put out about a year ago now,

and that this kind of really defines this research program that I've been working on for the last year and a half in some form or another,

which is looking at exactly this trade off using category theory theory.

So, here's a specification of a Turing machine.

This is just a simple deterministic computation.

It's saying, you know, you have a Turing machine that has this head state and this tape state,

and on the next step, you're going to replace the tape state with something that looks like this,

the head state with something that looks like that,

and you're going to scroll the Turing machine head left, or in this case, scroll it right, etc.

So, this is just a, you know, specification of a very simple computation.

I think this is a two-state, two-color Turing machine on a simple, you know, one-dimensional tape.

It's about as simple a computation as you could define.

So, if you run that thing for some initial condition, you'll get an evolution that looks like this.

And so, right now, this is just a purely deterministic, you know, single-path evolution.

But from this, we can construct, we can build a mathematical structure, namely, we can build a category.

So, and the rules for how we build that category are very simple.

So, you know, each arrow here is some simple computation,

some application of the Turing machine transition function.

And then what we can do is we can say, well, anytime we have two arrows that are laid end-to-end like this,

we can compose them together to create a third arrow that goes like that.

I may even have a picture.

Yes, like this.

So, you know, we have a computation F that takes us from X to Y,

a computation G that takes us from Y to Z,

and then we obtain a composite computation G compose F that takes us directly from X to Z.

And we also add some additional edges, some additional arrows on each state itself,

a sort of identity, an identity operation that maps a computational state directly to itself.

And so this, combined with some axioms of associativity and identity,

forms a category of elementary computations.

So this is a very, very simple example.

But what I want to try and build up towards and kind of pump your intuition for is a category which I call comp,

which is a category whose objects are essentially the class of all data structures

and whose arrows or morphisms are the class of all elementary computations.

So you start by just applying, you know, all possible computations or, you know,

in this case, for the case of Turing machines, all possible, you know,

Turing machine transition functions.

And then you do this closure operation where you, you know, where you essentially do what I'm doing here,

but, you know, you allow those elementary computations to be composed together in arbitrary ways.

And so that gives you effectively a class of all possible programs.

So this category comp contains not only all possible data structures as objects,

but all possible programs as morphisms.

And this is a very rich category with some very interesting algebraic structure that we'll kind of,

again, I'm sure we'll allude to in our subsequent discussion.

But in a sense, when we do this, when we do this operation of taking what mathematically we call a transitive closure,

right, where we allow two elementary computations to be composed together to produce a third,

we are essentially kind of neglecting considerations of computational complexity, right?

Because, you know, this arrow here might correspond to one application of the Turing machine transition function.

This arrow might correspond to another application of the transition function.

But this composite arrow might correspond to two applications of the transition function.

And so somehow when we allow arrows or morphisms to be composed in this way,

we're neglecting considerations of the complexity of operations.

So the question then is, you know, could we imagine constructing a generalization of category theory,

which takes into account computational complexity?

So here's an example of how that would look, right?

So here you can see every edge, every morphism has been tagged with certain computational complexity information.

In particular, it's been tagged with a number specifying what is the minimum number of applications of my transition function?

What's the minimum number of elementary computations that I need in order to evolve from this data structure to this data structure?

So here, to go from here to here, it's just one.

To go from here to here, it's one.

To go from here to here, it's one, etc.

But to go from here to here directly, it would be three.

To go from here to here directly, it would be two.

And, you know, just for convention, we say that the identity computation, the trivial computation,

always has complexity zero.

So this is, again, a fairly simple mathematical structure.

And you can construct this, again, using purely category theoretic technology

by building a particular functor from the category of computations

and from the category of data structures and computations

to what's called a discrete cobordism category.

And again, we might discuss that later on if people are interested,

but let me not get too bogged down into the technical details of how we do that.

But once you've got this, it gives us immediately a very nice way of characterizing

phenomena like computational irreducibility.

So there is this idea that has existed in some form or another

since the very early days of theoretical computer science,

since the days of, you know, Gödel and Turing and Post and Church and so on,

but was given this term computational irreducibility by Stephen Wolfram,

where the idea is essentially that you just, you know,

intuitively you describe a computation as being irreducible

or, you know, the result of the computation as being irreducibly complex

if it's not possible to shortcut it in any way, right?

So where, you know, it takes, that computation takes a certain number of steps

and there does not exist a shorter computation

that would give you the same answer in less time.

And one of the nice features of thinking about computations
and their complexity algebraically like this
is that it gives you a purely algebraic characterization of irreducibility.
In particular, what it says is that irreducible computations
are ones for which the computational complexity acts
additive, purely additively under composition.
So if it's the case that if we compose, say,
two computations of complexity one together,
if the resulting composite takes, you know, has complexity two,
then it's an irreducible computation.
If it has complexity less than two, like one,
then that means that we could have jumped directly
from the input to the output
without having to pass through the two elementary computations
that made it up.
So that would be an example of a reducible computation.
So reducible computations are ones whose complexities
compose sub-additively in this category theoretic sense.
And, okay, so here's an example,
here's an illustration of showing
what intermediate computational states you had to go through
in order to get from one data structure to another data structure.
So to go from here to here, you had to go through steps one to two.
To go from here to here, you had to go through steps one, two,
and one, two, three, and four, et cetera.
So you can build up a kind of complete algebra of complexity this way,
which has some nice properties, which again, I can talk about,
but let me not get too bogged down in mathematical details right now.
But here's the thing I really want to talk about,
which is what happens when you go to multi-way systems,
what happens when you go to non-deterministic computations.
So now imagine having, instead of just a Turing machine
with a single rule, a single transition function
that just evolves deterministically with a single thread of time,
now imagine having a Turing machine that has, say,
two transition functions, like this one and this one.
And so at any given point, it can apply one of the two.
And so now evolution, instead of just being a single path,

becomes this kind of branching structure,
which if we didn't have any merging would be a tree,
but because we are merging equivalent states,
it's actually just a kind of more general directed graph.
And so it looks like this, and this we call a multi-way system.
And so we can build a category out of these multi-way systems as well.
We can build a category using exactly the same rules.
So again, we do this transitive closure operation.
So we add an edge for every possible composition
of these elementary computations and an identity edge
that maps every data structure to itself.
But it turns out this category has even more structure
than the single-way system that we showed previously,
because now it's possible to compose computations
not just sequentially in time using ordinary composition,
but it's possible to compose them in parallel
across what is sometimes referred to as branchial space.
So essentially you're saying,
instead of, you know,
the ordinary morphism composition that I showed previously
is essentially saying, you know,
I apply this elementary computation,
then this elementary computation sequentially.
Whereas this parallel composition is saying,
I apply this computation and this computation
in parallel to the same data structure.
And so that parallel operation
is what causes these branches, right?
Effectively, when you have two threads of time
that are branching from the same state, like here,
that's arising because we have chosen
to apply this elementary computation
and this elementary computation together in parallel
rather than sequentialized in time.
And so here you can see this parallelization
indicated using what's referred to
as a branchial decomposition,
which is just a kind of a visual way

of decomposing what's going on
between these different threads of time.
And again, there's a purely algebraic characterization
of what's going on here,
which is that what we've done
is we've taken our simple category
that we started with
and we've equipped it
with a tensor product structure.
And so it's become what we fantastically call
a monoidal category
or actually a symmetric monoidal category.
So the tensor,
so we now have these two operations.
We have sequential composition in time
and we have this tensor product operation,
which is a parallel composition
in branchial space.
And just like we can have,
just like before,
where we equipped our edges,
our morphisms
with certain computational complexity information,
we can do the same thing
and we described
how those complexities composed sequentially in time.
We can do the same thing
and describe how the complexities compose in parallel
as one composes morphisms in branchial space.
And so this allows one,
by exactly the same token,
to quantify multi-computational irreducibility
rather than just computational irreducibility.
So now multi-computational irreducibility
becomes a measure
of how additive or sub-additive
your time complexities are
when you compose them

in parallel through the tensor products
rather than just in sequence
through standard morphism composition.
Okay, but I promise I am,
and so here's an analogous diagram
to the one I showed before
showing all the kind of intermediate steps
that are being applied
when one composes computation,
when one constructs computations
or indeed multi-computations
by composing elementary computations
both sequentially in time
and in parallel in branchial space.
Now, but I promise I am,
the point that I'm trying to get to
is that it turns out
that in addition to just being a useful way
to think about computational complexity theory
and to formulate complexity classes
like, you know,
polynomial time
or non-deterministic polynomial time, etc.
It turns out this is also an interesting way
to think about the role of observation
in sort of computational models of reality.
Because, so,
and here's where I'm going to get
a little bit philosophical
and I don't have an,
I don't immediately have a slide
or a graphic that I can show
to illustrate this point.
But so when we think about
modeling a system computationally,
one has to bear in mind
that there are really two computations going on, right?
There's the computation

that the system is itself performing
and then there's the computation
that the observer,
the person who is measuring that system
and concluding things from it,
there's the computation
that they are performing.
And somehow, you know,
so when we construct models of reality
or when we construct models of systems,
you know,
and we want to describe
kind of at a meta level
what we're doing in computational terms,
there's our own computation
that, you know,
that's going on
inside our own internal representation
of the world.
And then there's presumably
some external computation
that's going on outside.
And then when we make observations
and when we make measurements,
when we construct theoretical models,
what we're doing
is we're somehow constructing
some kind of encoding function
that allows us to take
a concrete physical state
of the system we're observing
and encode it
as some abstract state
of the internal model
that we have
of what's going on.
And that's all very well.
But then now,

we don't just have
one computation to care about.
We have three, right?
We've got the computation
of the system,
computation of the observer,
and the computation,
this encoding function computation
that's responsible
for their, you know,
the interface
between their internal model
of the world
and the external reality.
And the computational complexities
of these computations
interplay
in an extremely interesting way.
And so the, you know,
part of the reason
for trying to develop
this algebraic semantics
for thinking about
computational complexity
and multi-computational complexity
was to try to give one
a systematic way
to reason about
exactly this three-way interplay
between systems,
observers,
and encoding functions.
And so, in particular,
when we make,
when an observer
makes a model of the world,
one thing that they're doing
is that they are,

you know,
for any model
isn't just, you know,
a complete description
of reality,
there's a certain amount
of coarse graining, right?
There's a certain amount
of taking a bunch of states
that in the system itself
are distinguished,
but in the internal model
are treated as the same.
They're kind of,
you know,
they're cast in the same bucket.
So in some sense,
you know,
how coarse a model is
is a measure of how much
the encoding function
fails to be subjective, right?
And so, again,
there's a kind of algebraic
or category theoretic
characterization
of what's going on
that, you know,
the fewer of your morphisms
are epimorphisms,
the more coarse your model is,
you know,
the more abstract
or idealized
your model of reality is.
And so then,
the interesting thing
is that this characterization

of multi-computational
irreducibility,
this measure of how
additive or sub-additive
your complexities are
as you compose them
together in parallel
gives you a measure
of the relative complexity
of the evolution function,
that is,
the function that evolves
your computation forwards
in time,
versus the equivalence function,
that is,
the function that declares
that two computational states,
two data structures
are to be treated
as equivalent.
And that interplay,
I claim,
is a kind of abstract
meta way of thinking
about the interplay
between the computation
of systems
versus the computation
of observers.
Because, you know,
it says the role of the system
is to evolve forwards
in time,
whereas the role
of the observer
is to take states
in the, you know,

in the system
that are distinguished
in reality
and say,
you know,
subject to my idealized model,
I'm going to treat
these as the same.
So, you know,
the system is defining
the evolution function,
but the observer
is defining
this equivalence function.
And so then
the trade-off
in their complexities
becomes exactly
a trade-off
between how,
you know,
what are the,
what are the algebraic rules
that describe the complexities
as they compose sequentially
versus the algebraic rules
that describe the complexities
as they compose
under this tensor product operation.
And so, you know,
I've shown this in particular
for Turing machine systems,
but this is a very,
as I say,
this is a very general
kind of algebraic semantics.
You can apply it to hypergraphs.
You can apply it to combinators,

lambda calculus.
It doesn't matter.
In a sense,
there is just one category
up to isomorphism
of computation,
of data structures
and computations.
And there are simply
many different ways
of parameterizing
what that category is doing
through things like
Turing machines
or hypergraphs
or whatever.
You know,
the algebraic formalism
transcends
the particular details
of the computations
that one's dealing with.
And so, yeah,
as I say,
what one ends up with
is, I think,
a fairly general formalism
for thinking about
the interplay
between observers
and the systems
that they observe.
And that gives one a,
I promise,
I'll stop monologuing
in a moment
and we'll try and
pick apart

what I'm really
talking about here.
But so I'll just conclude
with, you know,
once one has
that algebraic semantics,
a whole bunch of things
which I think
previously would have been,
at least to me,
previously seemed like
kind of fundamental confusions
about, you know,
how scientific observation works
and how it interplays
with computational models,
those confusions
kind of become
much easier to clarify
once you think about it
in this kind of
more compositional way.
So to give a very
simple example
or a kind of
very degenerate example,
you can, you know,
within this algebraic semantics,
you can effectively
trade off
the computational complexity
of the system
for the computational complexity
of the observer, right?
So you can have
kind of, in effect,
two degenerate cases.
You can have the case

where the system itself
has a completely trivial
evolution function.
The system itself has,
you know,
is doing something
completely elementary
in how it evolves.
But then the observer
has some incredibly
complicated equivalence function
that makes the system
look like it's doing
something really complicated,
even though what it's
actually doing
is something very simple.
And so then you have
the phenomenon where
actually kind of
all of the complexity
is in the eye of the observer.
You can also have
the other degenerate case
where the observer
is doing something
absolutely trivial,
where the, you know,
the encoding function
or the observer's
own internal representation
is just an identity
function or something.
So there's no complexity there.
But the system
is doing something
incredibly complex.
It's doing some totally,

some really sophisticated
universal computation.
And so that will also
appear very complex
to that observer.
And so,
and you can also have
any kind of,
any intermediate,
you know,
there's this vast
interstitial space
between these two extremes.
And so one thing
that's kind of always,
one sort of philosophical problem
that I've always kind of
been interested in
ever since I was a kid,
which is this sort of,
this tension between
empiricism versus rationalism,
right?
You know,
the question of,
you know,
on the way,
if you look back
at the history of,
you know,
early European philosophy
or, you know,
certainly Western,
you know,
Western post-enlightenment
philosophy,
you had people like,
you know,

Descartes and Leibniz
and so on who were,
you know,
and in a more sophisticated way,
Bishop Barclay
with subjective immaterialism
who were trying to push
for this idea that,
oh, you know,
all the sophistication
is what's going on
inside the observer's head
and, you know,
what goes on in reality
is somehow secondary.
And then you had people like,
you know,
Locke and Hume
and the empiricists
who are saying,
no, no,
we should try and get the observer
as much out of the picture
as possible
and we should say
all the sophistication
is going on
kind of in the external world.
And this,
you know,
one nice consequence of this
is it gives one,
one nice consequence
of this formalism
is it gives one actually
an algebraic way
of kind of parameterizing
this spectrum

from rationalism
to empiricism,
right?
That you can choose
the rationalist extreme
where, you know,
you know,
you just have some
space of all possible computations
and the observer
is doing all of the work
to try to narrow down
to a particular one
or you can have
the kind of empiricistic stream
where, you know,
the observer is
a completely elementary system
and, you know,
and everything they observe
is just being built up
from a kind of bottom-up
construction
or you can have
anything in between
and in a sense
we now have,
I think,
the beginnings
of a mathematical theory
that explain,
that's able to explain
how those complexities
trade off
in a very direct way.
So I think there's
potentially places
of mutual interest

there in kind of the,
in thinking about,
yeah,
as I say,
cognition,
observation,
measurement,
scientific modeling
and so on
in fundamentally
computational terms.
So I think that
hopefully that will
provide some context
for a discussion.
Thank you.
Great opening.
There's so many places
to spin in
and jump through.
I guess I'll start with
the two things
I wrote down
were unity is plural
and at minimum two
and beauty is in the eye
of the beholder
and the way that
these kinds of pieces
of timeless wisdom
that describe
that fundamentally
relational component
to observers
in systems,
which are not
all kinds of systems
per se,

but those kinds of systems,
those things are true for.
And then the way
in which along
the formalism
that you described,
with the system
observer encoding
three-way partition
and then the way
that in free energy
principle
and the particular physics
that interface
gets broken out
from the agent's
perspective
into the incoming
sensory
and the outgoing
action.
So then that results
in the four-fold
particular partition.
So maybe just
to kind of
dwell there,
how do we partition
the,
from a category
theory perspective
or however,
the action
perception loop
or the engagement
loop?
Like,
how do we

make a topology
or compare
or contrast
different topologies
and flows
over this kind
of seemingly
pervasive
or universal
interface-like
concept?
That's a fascinating
question.
So I don't have
the answer.
This is maybe
a place where
both you, Daniel,
and perhaps David
may have useful
perspectives on this
because,
you know,
I'm,
you know,
I read Carl's work
a few years back
and so I have some
familiarity with the terms,
but I'm by no means
a kind of,
you know,
an expert on free
energy principle
or, you know,
active inference
and those kinds of things.
But I think it's a very

good point that you raise
and so,
you know,
I should begin by just
being honest and say
that, you know,
everything I'm,
you know,
all that I just described
is, of course,
an idealization
and that,
you know,
in reality,
you know,
in particular,
it's an idealization
which I think you were
very right,
Daniel,
to kind of pick up on.
It's an idealization
in which we say
the observer is completely
kind of non-interacting
with the world somehow,
right?
That, you know,
in a sense that
there's just input coming in
and nothing coming out.
But of course,
we know that's not
really how observation works.
Observation is necessarily
a kind of two-way process
and so what's needed
is not just this kind of

very clean algebraic semantics
that I've described here
which assumes that
there's essentially
a one-way function
from the world
to the observer
but actually something
more like a kind of
second-order cybernetics
description of,
you know,
of what's really going on
where you have,
you know,
first-order
and second-order
interactions
where it's exactly
as you say
you can get these
sort of feedback loops
from observation to action
and back again
which are probably,
which is,
I mean,
still an idealization
but probably a more
realistic idealization
for how real observers
and real measurement apparatus work.
So,
I just want to begin
by saying that
I don't know,
right,
and, you know,

the question of how
this formalism interplays
with things like
second-order cybernetics
and other areas
where I know
these kinds of questions
have been explored,
that's something
I'm very interested
to find out about,
you know,
going forward.
but I think,
hmm,
okay,
so,
yeah,
so at some point
maybe you could
help me understand,
you know,
potentially where things
might fit in
with the,
you know,
with the kind of
broader active inference
framework.
I'm not sure
I necessarily have
that much more
to comment on
than that.
yeah,
other than to say
that,
you know,

in a sense,
okay,
so maybe,
you know,
one further comment
is that,
because,
I mean,
you asked specifically
about how the kind
of compositional
category theoretic
perspective might be useful.
So,
as I said,
I don't think category theory
in itself is going to be
the complete answer.
I think it will be
category theory augmented
with some other things,
computational complexity,
probably second-order
cybernetics,
and some other things
that I may not be aware of.
But one place
where I think
that viewpoint
is useful,
at least on a
philosophical level,
is the idea that,
is the idea that comes
about,
that you really
obtain by studying
mathematical structures

in a category theoretic way,
which is that
the identity of something,
you can define it
both in terms
of its intrinsic properties,
or you can define it
in terms of,
you know,
the stuff that you
can do to it,
right?
So,
you know,
this was really
the transition
that happened
in the foundations
of mathematics
as a result of people
like Samuel Allenberg
and Saunders-McLean.
So,
you know,
category theory
has its origins
in this sort of
slightly abstruse branch
of algebraic topology.
It was,
you know,
initially developed
by people like
Alexander Grotendieck
and Jean-Pierre Serre
for doing homological
algebra for,
you know,

for reasoning
about sort of
the algebraic structure
of topological spaces.
But then later,
in the,
I think,
1960s,
1970s,
these two American
mathematicians,
Eilenberg and MacLean,
realized that it was
useful not just for
thinking about topology,
but for thinking about
kind of mathematical
structure in general.
And then later on,
applied category theorists
started saying,
well,
maybe it's useful
for just thinking
about structure in general.
But,
you know,
the key kind of
conceptual or philosophical
shift that it imposes
is,
you know,
historically,
thanks to the work
of people like
Cantor and Frege
and Russell and so on,
people have thought

about mathematical structures
and the foundations
of mathematics
as,
you know,
in terms of set theory.
And the idea in set theory
is you have this,
you know,
things like the axiom
of extension
that effectively say
set is defined
by what's inside it,
right?
So in other words,
you know,
you,
you,
you,
a mathematical structure
obtains its identity
by,
you know,
you break it apart
and you look at
what's inside.
In category theory,
it's a completely
different view.
The view instead
is you say,
well,
no,
you can't look inside,
you know,
it's a fundamental rule
of category theory

that you can't look
inside an object.
You know,
its internal structure,
if it has any,
is sort of out of bounds
to you.
And instead you,
you give that object
identity in terms of
how it relates
to other objects
of the same type,
right?
So in other words,
you know,
you can ask,
you know,
what,
what can I do to this?
What,
what functions can I apply to it?
What functions can,
can I apply to something else
that map into this?
So,
you know,
if I want to define,
I don't know,
the real numbers
or the integers
or something in the set theory,
you know,
from a set theory perspective,
you would say that the,
you know,
the,
the,

the essence of the real numbers
are all the numbers
that are inside that set,
whereas the category theory
perspective is no,
the essence of the real numbers
are all the functions
that you can define
that take real numbers
to some other number system
or real numbers to themselves
or that take some other number system
into the real numbers
or et cetera.
And,
you know,
some of the deepest results
in category theory,
like the Ornade dilemma
and other things
are telling one
in some very precise sense
that these two perspectives
are really the same
at some fundamental level,
but that,
you know,
identifying an object
based on its internal structure,
based on breaking it
apart and asking
what's inside
and identifying an object
by asking
what can I do to it
and what,
you know,
what can this object

be transformed into
and what things
can be transformed
into this object,
those give you
exactly the same information.
It's far from obvious
that that's true.
You know,
the Ornade dilemma
is a very kind of,
it's one of those results
where you can never
quite work out
if it's obvious
or if it's incredibly,
you know,
mysterious,
but I tend to fall
on the side
that it's incredibly mysterious.
It's far from evidence
that those two perspectives
would really be the same.
And yet,
the point you're making,
Daniel,
I think,
is that,
in a sense,
historical ways
of thinking
about scientific observation
have tended
towards the set
theoretic viewpoint,
they're tended
towards the viewpoint

that we understand
systems based on
kind of breaking
them apart
into constituent components.
But perhaps a more
realistic view
is something more
like the category
theory perspective
where we say,
you know,
I understand the system
by interacting with it,
right,
by asking what can I do to it
and how does it behave
when I perform
certain operations to it.
And that's a fundamentally,
you know,
that's a fundamentally
two-way process
that involves
not just passive observation
but also kind of
active participation.
And somehow,
we need to develop
a formalism
that kind of
incorporates those two elements.
And maybe,
you know,
maybe it already exists
and it exists
in this large literature tree
of which I'm largely unaware.

That's partly why
I want to be here
to try and find out,
you know,
what things I missed,
so to speak.
Excellent.
All right.
A few points.
Self-evident
is far from evident.
Also,
I tend to the mysterious
which is to say
saying more with less,
especially for these frameworks
because they're less opinionated
so that their space
of internal semantics
can be larger.
And then
that description
that you provided
with the relationship
between the set
and the category theory,
so I kind of
summarized it as
set is to essential
inclusion
as category
is to relational function.
Now,
if our concept
of organismality
or of action
in the niche
is constructive

compositional material,
then we are looking
for like
what is in or out.
Is the microbiome in?
Is the pheromone
in the ant colony
in or out of that thing
because it's looking
for like a static
material answer?
And then,
in contrast,
the other side
of that coin
highlights the dynamic,
like whatever it is
that's self-organizing
of the tornado
is the tornado.
Whatever it is
that's self-organizing
for the ant
is the ant.
and then also
this like
hint slash
Mobius strip
or something
that those two
in the moment
are indistinguishable
and yet systems
that we choose
to define one way
or another
or keeping both open,
those design decisions

do make all the difference
even if for real systems
as they're observed,
there's indistinguishability.
Right, right.
I think that's
a very important point
and one which,
I mean,
this is a,
it's always a kind
of concern I have
whenever I start thinking
about embodied cognition
or extended phenotype
type ideas,
right?
That,
in a sense,
if what one is trying
to do here
is construct
a kind of formalistic
model of observation
or of cognition
or something,
then as a kind
of first order
approximation,
one has to start
by somehow decomposing
the world
into observers
and systems.
But of course,
we know that
that decomposition
is somehow

arbitrarily imposed,
right?
And that,
you know,
if you take these things
to their extremes
and you allow
essentially everything
that the agent
is interacting with
to be considered,
like not just
their microbiome,
as you say,
but also tools
that they construct
or environments
in which they exist
and so on.
If you,
as you start
to consider all of that
to be a component
of that organism,
you know,
of that agent's phenotype,
which is a completely
reasonable thing
to do,
then,
and you start
to,
you know,
you start to say,
okay,
well,
their cognitive processes
are not just localized

to their brain
or their spinal column,
but are kind of
somehow extended
to,
you know,
the computers they use,
the paper they write on,
the books they read,
et cetera.
Again,
perfectly reasonable
thing to do
and sort of somehow
more descriptive
of what's really going on.
But my fear is always
if you take that too far,
then,
you know,
you end up destroying
the whole assumption
that the idealization
was based on,
which is that you can
neatly decompose,
you know,
the world into observers
and systems.
And so I always get a bit
nervous when thinking
about that,
that it's like,
yes,
you know,
in a sense,
yeah,
we know this is an approximation

and we know that
that approximation
is not really true,
but how,
you know,
how much can you afford
to sort of
loosen your grip
on that approximation
before the whole thing
just kind of falls apart?
I don't really know
the answer to that question,
but I think it's
an interesting one.
Yeah.
How about I'll ask
some questions from chat
and then give your first thoughts
and then we'll see
maybe where that kind of
lands with further questions
or how it connects
to active inference.
So that sounds good.
Oh,
by the way,
should I keep my screen share on
or should I say?
Yeah,
we might want to go to a figure,
so it's fine.
Okay.
Yeah.
All right.
Quantum Bell wrote,
how does this help us
reason about causality?

That's a fascinating question.
Okay.
So that's another major aspect
of, you know,
why I think this research program
is exciting because,
so,
and again,
this is something
where I'm interested
to get the kind of
active inference perspective
because I know this,
again,
it's a topic
in which much has been written
and I'm largely ignorant.
But,
yes,
so one question you could ask is,
yeah,
if you have a description
of a computation like this,
like,
let's go back up
to the Turing machine case,
the single way Turing machine case,
that's relatively easy to analyze,
although still far from obvious
what's going on.
So suppose you have
a computation of this kind
and you want to ask,
what is its causal structure?
In other words,
you know,
for each edge,
each time I'm applying

this Turing machine
transition function,
can I construct
some kind of graph,
you know,
some directed graph representation
that tells me
how these events
are linked together?
So in the,
within the original
research program,
this so-called
Wolfram model research program
that kind of started
a lot of these investigations,
we were looking at this
all the time,
right?
We were looking at kind of,
you know,
taking computations
and looking at that causal structure
and trying to,
you know,
infer things about
what was going on
about the,
you know,
the semantics of the computation
based on causal relationships.
And at a certain point,
I started to realize,
and I think other people
had realized this before I did,
but I'm often slow
to pick these things up.
I,

I,
I,
I and other people
started to realize
that the notion of causality
we were using
was kind of nonsense.
I mean,
it was not completely hopeless,
but it wasn't really causality
or it couldn't really be called causality
in any,
in any definite sense.
So what do I mean by that?
So first of all,
it was a very technical problem.
So if you're looking at something
like a Turing machine evolution
or a hypergraph rewriting system
as we were,
then there's a very tempting
and apparently obvious
natural definition of causality
that you could use,
which is to ask,
you know,
when you split the world up
into events that take,
you know,
some part of your data structure
as input and output,
you know,
some other part of a data structure
as output,
then you can very easily ask,
well,
does the output of one event
intersect with the input

of another event?
So if I show the hypergraph example,
it's perhaps easier to see.
So you have a hypergraph rewriting rule
that looks like this,
right?
So,
you know,
you have,
you say,
if I have a piece of hypergraph
that looks like that,
I replace it with another piece
of hypergraph that looks like this.
So at each,
each time you apply an event,
you can think of that event
as,
you know,
ingesting certain hyper edges
and kind of,
you know,
replacing them with others.
So you can divide it up
into a sort of the,
the input hyper edges
that are being ingested
versus the output hyper edges
that are being produced.
And so then you can ask,
well,
do the output,
did I use,
did I subsequently ingest
in some future event
hyper edges
that were output
in some previous event?

Well,
if the answer is yes,
then pretty obviously
that future event
couldn't have occurred
unless the previous event
had already occurred.
So then you could say,
well,
then one of those events
causes the other.
So in general,
you could say that two of it,
you know,
event A causes event B
if it's the case
that the output,
that the,
the collection of tokens
that was produced
in the output of event A
has a non-zero intersection
with the collection tokens
that were ingested
as part of the input
of event B.
And that's a very tempting,
very natural definition
of kind of causality
in these systems.
Turns out it doesn't really work.
I mean,
it works pretty well,
but there are cases
in which it fails
and it fails pretty spectacularly.
And so the kind of canonical case
where it fails spectacularly

is that you can have events
that don't actually do anything,
right?
You can have events
that just kind of touch an edge,
touch a token
and output it again unchanged,
but maybe,
you know,
change the name.
It modifies the name.
It modifies the identifier,
but it doesn't actually change
anything about the structure
of the hypergraph
or the Turing machine state
or whatever.
So pretty obviously
that event doesn't matter.
It can't,
it shouldn't be causally related
to anything in the future,
but because it modifies,
because it ingested the edge
and then didn't do anything,
you know,
did some identity operation,
but then,
you know,
produced it in the output again,
it will kind of register
as being causally related
to any future event
that used that edge,
even though it didn't
make any difference.
That's just one
very obvious example.

There are other cases
where it became clear
that whatever this thing was,
whatever this algorithm
was detecting,
it wasn't really causality.
So I tried to think about,
you know,
what's a more sensible
definition of causality,
and I started working
on things to do with,
you know,
a slightly kind of
blockchain inspired ideas
where you say,
okay,
well,
rather than just arbitrarily
assigning,
you know,
identifiers to these tokens
every time they're created,
what if I recursively
construct the identifier
of the token
based on its causal history?
So in other words,
each token,
like each hyper edge
or each state in my,
each tape square
on my Turing machine tape,
the identifier is not just
some random number
that gets generated
by my algorithm,
but instead,

its identifier
is a directive graph
representation
of its complete causal history.
Well,
then kind of recursively,
its identifier
can only change
if the causal history
was updated,
and so you don't end up
with these kind of
spurious causal relations
that I described before.
So that seemed like
one tempting way
of resolving this problem,
but then I realized,
actually,
there's a much more
fundamental problem.
There's a philosophical problem
with the way
that we're thinking
about causality,
which is that
it's not really,
you know,
so,
okay,
this is a long tangent,
which I'll talk about
a little bit,
but I won't get into
the complete details
unless people are interested,
but I ended up talking
to a bunch of philosophers

who,
you know,
who worked on causality
and people who worked
on parallel programming
and quantum information theory
and other places
where causality was studied
and asked them
kind of basically,
what do you mean by causality?
What is causality?
What is this thing
we're trying to define?
And in some form or another,
all of the definitions
boil down to,
you know,
event A causes event B
if,
had event A not occurred,
then event B
would not have occurred.
So in other words,
you need a counterfactual.
You need some possible history,
some possible world
in which event A
didn't happen.
But if you're reasoning
about a purely deterministic
event system,
like a Turing machine,
that doesn't make any sense.
Because if you're,
you know,
if you have a single
Turing machine transition function,

there is no possible world
in which that transition function
didn't fire in that particular way.
Because if it didn't fire
in that particular way,
you would not be reasoning
about that Turing machine anymore.
You'd be reasoning
about a different Turing machine.
So suddenly this,
you know,
to make sense of these notions
of causality,
you need a kind of
Leibnizian,
you know,
modalities view of reality
that just doesn't exist
for these deterministic
computational systems.
So either you need
to define computation,
you need to define causality
only at the multi-way level,
only at the level
where you have many computations
or possibly all computations
happening in parallel,
and then you can define
causality relative to all of,
you know,
relative between them.
Or you were kind of hosed,
right?
There wasn't really,
you know,
that seemed like the only kind
of,

the only get out.
Or you need some fundamentally
new philosophical theory
of causality
that I was not qualified
to produce.
And so,
so that's,
again,
part of the reason
what,
part of what motivated
this general research program,
which is trying to think
about this category
of not just a single computation
and with a single sequence
of data structures,
because it's clear
that you can't,
in a philosophically meaningful way,
assign causality in that case,
but rather,
you know,
looking at the algebraic structure
of the category
of all possible computations
and all possible data structures
and in that situation,
there is a notion of causality
you can equip that with.
And there's a nice,
again,
a nice mathematical description
in terms of,
in terms of weak two categories
and so on,
which again,

I can,
I can talk about
if people are interested.
But,
but yeah,
so it's clear that these things
are very deeply related,
that this,
this sort of theory
of the category of computations
and data structures
and the theory
of how you assign causality
in a meaningful way
are very deeply related.
And I'll just mention
one other thing on that topic,
which is,
again,
just an area
which I find
quite exciting
because it's,
it's an unexpected spin out
of this program,
which is that,
so once you have a way
of consistently applying causality
at a per token level
in these systems,
it gives you a way
of vastly generalizing
what computation is.
And you can,
in particular,
you can get,
you can derive something
which I call,

well,
which I'm provisionally calling
covariance computation,
although it should probably
have a better name than that,
which is,
so,
you know,
in our traditional
kind of Turing-Church
type,
you know,
type models of computation,
computation is a purely
forwards-in-time operation,
you know,
so at every point,
you know,
you have a complete data structure
and computation is about deriving
what is the next state
of that data structure.
So in a sense,
it's only a forwards-in-time thing.
You might be able to kind of
reconstruct the initial conditions
based on some subsequent
data structures,
you might be able to go backwards
in time,
but that's essentially
what you're doing.
But then,
you could imagine,
okay,
suppose I don't know
the complete state
of my data structure.

I know,
instead,
I know one part
of my data structure,
but I know it,
I know its history
throughout all of time.
So you could imagine,
say,
an elementary cellular automaton
or a Turing machine tape
where you know nothing
about the tape,
but you know the state
of one cell
and you know it,
you know,
throughout all of time.
And then the question is,
what can you infer
about the rest of the computation?
And it turns out that
for those kind of structured
array type systems,
you can infer a lot.
You can actually evolve
the system,
not forwards in time,
but sideways in space
and obtain a kind of
causal diamond that,
so,
okay,
the top left,
top right,
bottom left,
bottom right corners
are undetermined,

but everything inside
that diamond can be determined
just from that one row
or that one column
that you know,
you know,
sort of extended
throughout time.
And so,
you know,
and that's a fundamentally
different notion of computation.
So it's a version of computation
which is not forwards in time,
but sideways in space.
But you can also have,
version of computation
that are sideways
and branchial space
where you know,
you know,
one complete state,
you know,
you know one branch
of the multi-way system
extended throughout time.
And then the question is,
what else can you infer
about the,
you know,
but the rest of the multi-way system
just from that one branch?
And again,
the answer turns out to be,
you can infer a lot,
but not everything.
And so just like in,
the reason I call this

covariant computation
is because it's very analogous
to what happens in relativity.
So in relativity,
once you buy into this notion
of general covariance
and the notion that space
and time are kind of
fundamentally the same thing,
then you have to somehow
relax your traditional view
of what dynamical systems do,
which is,
you know,
we typically think of systems
as evolving,
you know,
you have a snapshot
of your data
at,
you know,
localized on a,
you know,
for a particular state of space,
you know,
on a particular space
like hypersurface.
And then your laws of physics
tell you how that space
like hypersurface evolves
forwards or backwards in time.
But in a covariant picture
of physics,
then you must also allow
for,
you know,
your initial data
to be defined

on a time-like hypersurface
and for you to be able
to evolve that time-like hypersurface
sideways in space
or mixtures of the two
and so on.
And so it's clear
that there's a very general,
a vast generalization
of ordinary computation theory
that you can,
that you can construct
that's kind of physics inspired
in that sense.
In which you can have
mixing of space,
time,
and kind of multi-way directions
in a completely consistent way.
But to make those things consistent,
you need to have a definite way
of assigning causality.
You need to,
because,
you know,
any computation that you do,
even if it permutes
the directions of space
and time
and branchial space
and so on,
must always somehow
preserve the causal structure.
It has to respect
the causal structure
of what's going on
or else it's inconsistent.
And so this question

of how you construct
a covariant theory
of computation
is,
it turns out,
intimately related
to the question
of how you take
this category
of computations
and data structures
and equip it
with a consistent notion
of causality.
So,
very interesting question.
We could talk about
that at great length.
Okay.
To follow with a few pieces,
it's very related
to Professor Mike Levin's
notion of polycomputing
and about the necessity
for a causality concept
to be created
or deployed
when the question arises,
was that me?
Was that action
or change due to me?
Also,
connectivities,
even just in the
neuroimaging setting,
which is kind of
the cradle
from which active inference

and free energy principle
arise from,
it's really important
to distinguish
the functional,
effective,
and anatomical
connectivities.
And that was one
of the points
that Toby Sinclair-Smith
made in his dissertation,
which is that
a lot of times
the Bayesian graphs
don't convey
all of the necessary
and sufficient information
to make the
reproducible computation,
which is one of those
kind of,
what's missing from the graph
is what motivated
a lot of the category theory
developments
in active inference,
as well as some of the
formal ontological works
with Sumo
and Dave here
and Adam Peace
because implementing
modal and higher order
logics is really important
if it's a possible situation
where a mind
can have a perspective

on a mind
and all these things
like that.
Then the
ant Turing tape,
the tape is the pheromone
and then the decision space
is the nestmates
scrolling.
So when you had
a deterministic Turing tape,
that was like a movie
because the nestmate
couldn't make any choices
except for internal action,
which is kind of
side topic.
But it couldn't make
any choices on the tape.
Whereas when there's
a multi-way,
which is basically
in active inference,
what we talk about
in terms of affordances
and the policy space
and the temporal depth
of planning,
counterfactuals on action
and action conditioned
world transition states
like the B matrix,
all those kinds of topics
come into play
because if you want
to have a causal buffer
or grasp on
what is it that

something that could do
otherwise does,
what does it cause to do
when it does or doesn't
do otherwise,
you need something
like a deterministic handle
around what could be
a probabilistic
or deterministic
but at least multi-way
map of some kind
of cognitive territory.
That's a very,
very interesting perspective.
I mean,
so it's,
again,
I'm betraying my ignorance
of active inference theory here,
but,
you know,
it sounds almost like,
so these,
these,
when you have this
kind of interplay
between,
I don't know,
the,
you know,
between sort of,
you know,
epistemic versus,
is it pragmatic?
There's,
you know,
there's sort of two aspects

of,
you know,
of how,
how this kind of speculative
passive cognition works.
I,
this is something
which I thought about,
I mean,
I thought about
in a completely different context
in relation to things
like quantum information theory,
but I wonder if it's,
I wonder if there's
a potential overlap there
between,
I mean,
so there are certain
situations when thinking
about these kinds of systems
purely abstractly
where you kind of need
two different notions
of causality.
You need a kind of
speculative notion
that can be,
that's dynamic,
that can be rewritten,
and then you need
a kind of definite notion
that's,
that's immutable,
right?
And so,
you know,
a classic example of this is,

you know,
for something like
quantum information theory,
right?
So you can have
superpositions of causal orders,
you can have quantum switches,
you can have,
you know,
you can have,
causal structure
that exists in,
in,
in superpositions
of different kind
of directed graph states,
but then once you apply
Hermitian operator,
once you apply a measurement,
the causal structure
is definite
because then everything
is relativistic
and you have covariance.
And you have similar things
as I understand
with,
with,
with,
with distributed computing,
with parallel computing,
where you,
you,
you potentially allow
for speculative execution
for a certain number
of steps
where you're kind of

treeing up this multi-way
system and you have
a superposition
or,
you know,
at least a collection
of possible causal histories,
but then eventually
you have to choose
an actual operation to do
and then the causal history,
you know,
you have this big block
that gets laid down
and,
you know,
and then the causal history
is,
is somehow definite.
And I wonder
if there's a,
there's a way
of thinking about,
sort of,
you know,
you know,
speculative
execution of agents
and the interrelation
between that speculative
execution and,
you know,
and agent actions
in terms of,
again,
this interplay
between,
you know,

between two different
causal structures,
between a dynamic one
versus an immutable one.
Yeah.
Well,
one funny way
to think about that
is like a single agent
that has this
counterfactual
contemplative ability
could be
in the center place
foraging arena
and then imagining
with discrete
branching paths
like a chess algorithm
or like a
probability distribution
could be like
imagining where
it could go.
But not all
cognitive things
are the kind of things
that make plans
of their own actions
whereas like
an ant colony
has nest mates
on the ground.
So they're actually
realizing
in these finite
trajectories
the real

exoskeleton
on the ground
that plays out
ending up
with those
simulated trajectories
could have been
simulated
or could have been
probabilistically blurred
but that's kind of
the difference
between like
the embodiment
and like
the body moving
there for a mammal
or for an animal
and then like
the mind simulating it
and then just to the
um
the epistemic
and pragmatic
trade-off
in decision making.
So
let's just say that
we're in that
multi-way
moment.
We'll just have
two options
two different slices
of the B
variable
and the policy
selection question

is about
which way
are you going to go
which affordance
in the moment.
Policy is basically
the affordances
for the time horizon
of planning
but if it's only
one time step
or just the next one
then the affordance
space is just
the actions
that can be taken.
Um
one way
to
make it so that
what happens
is the likeliest thing
path of least action
which is kind of
what opens up
the whole physics
of cognitive systems
angle
in contrast
to like a
reinforcement
reward learning
perspective
what makes it
the likeliest thing
is starting
with habit
so it could just

be drawn from
a fixed habitual
distribution
however for
adaptive action
habit gets up
weighted with
expected free energy
which is a
functional that
takes in
the policy space
which is summing
up to one
because there's a
probability over
actions
and then
up weighting
policies
according to
their score
on expected
free energy
which is
consisting of
epistemic plus
pragmatic value
so how much
is it going to
align the
observations to
be what I
like to see
that's pragmatic
value with a
preference
what is my

expected information
gain
that's the
epistemic value
so how those
are parameterized
make the agent
that always
seeks out new
information
or always goes
with habit
there's so much
policy space
because the
knobs are
not just
simple sliders
or there are
multiple knobs
even though
they are seemingly
quite
consilient and
minimal
like it's hard
to imagine
less
yet especially
when there's
richness in the
environment
even simple
systems can
have like
enormously
complex or
adaptive

behaviors
I'll just leave
it there
no I never
really thought
of the
so okay
the yeah
two things
right so
first of all
the perspective
of you know
thinking of an
ant colony
or a termite
colony or
something as
being you know
akin to a
mind
that's you
know I was
familiar with
that perspective
from people
like Dan
Dennett and
so on but
the idea that
the individual
ants in that
colony are in
a sense enact
they're kind of
the hardware
enacting the
speculative

execution
that's a
very interesting
idea
it's not
speculative
for them
well yeah
no exactly
but it's sort
of from the
mind's perspective
I guess it's
almost like
speculative
execution but
it's speculative
execution that's
being actuated
in the physical
world which is
which is very
interesting I
yeah I not
really thought
about that
before
but then yeah
okay so
then the
yeah the
point you're
making about
connection to
free energy
and sort of
habit formation
and so on

okay so
I wonder if
so you know
if we're
thinking about
a you know
a model of
cognition in
which there
are these two
different you
know two
distinct causality
notions the
immutable versus
the dynamic
one I wonder
if the so
you gave a
very very nice
account of how
you know habit
formation sort
of works in
these kinds of
formalisms based
on you know
prior experience
of expected
you know free
energy so
I wonder if
there's a way
of describing
that abstractly
in terms of
something like
you know you

you perform the
speculative execution
step where
you know you're
treeing out
several multi-way
possibilities and
initially you
kind of you know
you know nothing
or you have no
habits you're just
kind of you're
treeing everything
out with kind of
equal weighting
but then you
know for each
possible part you're
calculating either
an actual or an
expected free energy
and then somehow
you know in
future speculative
executions you
weight those
paths which you
previously had found
to have higher free
energy as higher
and so you're you
know you're more
likely to explore
those and less
likely to explore
ones which are
you know which

have that lower
expected value
it feels like
something something
like that should
should fit very
very nicely into
an algebraic
semantics like
this which would
be interesting
cool how about
more questions from
the chat
okay upcycle club
writes acknowledging
the limitations of
traditional entropy
in multi-computations
motivates us to
develop context
specific entropy
metrics can you
share some insights
towards such
efforts
yeah I can
certainly try
so yes I mean
the first point is
that you know
it's I think it
was there's that
famous conversation
between John von
Neumann and Claude
Shannon where I think
von Neumann famously

said that like
Shannon should call
his measure entropy
because no one knows
what it means right
and I submit that
the reason no one
knows what entropy
means is because
it's dependent I
mean okay one of the
reasons no one knows
what entropy means is
because it's dependent
on exactly what we've
been talking about
it's dependent on the
equivalence function of
the observer so it's
one of these things
like I don't know
maybe this is a stupid
analogy to use but
it's sorry I'm I'm
going to go off on a
tangent but I promise
it's sort of relevant
but so one thing is
okay one thing that
always breaks my brain
is when I try and
think about like
actuaries and life
insurance policies
because it's one of
those areas where
those models only
make sense if

they're not perfect
in a sense right
like so if you
had an actuary who
knew exactly how
long everyone was
going to live and
somehow that
information was kind
of openly available
there would be no
like life insurance
policies would be
pointless it's but
you know whereas
also if you had a
model that was
completely hopeless
of predicting how
long people would
live they would
also be hope
they would you
know life insurance
policies will also
be pointless
that is somehow
the very existence
of actuarial science
relies on your
model neither being
perfect nor being
awful it somehow
has to exist
somewhere in between
and as I say
that's something
which I it's one

of those topics
where if I think
about it for too
long it all just
stops making sense
and entropy has
very much that same
character right
because if you if
you were Laplace's
demon if you you
know if you had
perfect information
about the system
that you were
observing there's no
notion of entropy
right it's just
every you know
every you know
every micro state
so that you know
so you know the
Boltzmann formula
gives you an entropy
value of zero
where but you know
the notion of entropy
only exists once you
take once you take
that that perfect
knowledge of a system
and you coarse grain
it you define as I
was describing earlier
you introduce an
encoding function that
is not 100%

subjective so that
now you are mapping
certain distinct
micro states onto the
same coarse grained
macro state and then
now you can ask
okay what's the
number of micro
states that you
know exactly how
coarse is my
coarse graining
what's the number
of micro states
consistent with this
macro state what's
the number of
different values of
my domain that gets
mapped to a single
point in my co
domain of my
encoding function and
that's what entropy
is and so it's it's
very closely I mean it
is effectively a
measure of how good
is my coarse
graining so if you
had perfect knowledge
there's no entropy
if you have no
knowledge there's no
entropy it relies upon
you having a a not
completely trivial but

also not 100% you
know a a slightly
subjective but not
100% subjective
encoding function just
like with life
insurance policies but
so the reason the
reason I'm stating
that is because so
now it kind of I
think from that
perspective becomes a
little bit clearer why
there are all these
different notions of
entropy and why as the
questioner was alluding
to why entropy seems to
be so as a concept
seems to be so domain
and system specific
because every different
system every different
observer will in
principle have a
different set of
encoding functions a
different set of
equivalence functions and
each one will give rise
to a different
calculation of entropy
and so one way that
you can think about
this program this
program to try to
understand the

algebraic interplay
between time
complexity versus kind
of equivalence in
complexity or or you
know computational
irreducibility
irreducibility versus
versus multi
computational
irreducibility in
some sense that is a
program to try to
understand how
different definitions of
entropy relate to each
other in these kinds of
systems and how you
know if I if I take one
idealized observer that
has this equivalence
function and I and I
ask okay suppose now
they communicate with
this different observer
with a different
equivalence function they
come to different
understandings of what
the entropy of the
system is but what is
the relationship between
their measured entropy
values there clearly is
one that depends
algebraically on some
details of the of the
distinction between their

respective equivalence
functions but there
doesn't seem to be any
yet any general theory for
you know for how those
things are related and
that you know that's
part of the kind of the
raison d'être of this
of this research
program but yeah I
mean okay so one thing
that I will comment on
although this is a
little bit more
speculative it's again
sort of quasi
philosophical comment
but so one place where
these notions of entropy
become one place where
the fact that you have
all these different
notions of entropy
becomes kind of
interesting is in
fundamental physics so
when you start to think
about if you try to
model physics and the
universe in these
fundamentally computational
terms then one fairly
generic sort of
conclusion that you can
reach is that
gravitation general
relativity is essentially

an entropic
phenomenon I mean
Valinde and people
have kind of talked
about this in in in
non-computational
contexts too but it's
very very natural if
you start to think
about space-like
hypersurfaces being
sort of hypergraphs
then you know in order
to obtain a continuum
geometry that's
compatible with the
Einstein equations you
need to have certain
ergodicity you know you
need to be able to make
certain ergodicity
assumptions on the
rewriting which in turn
sort of implies certain
lower bounds on the
entropy of the system so
somehow gravitation general
relativity is a is a
coarse-grained theory you
obtain in the limit as the
entropy goes to infinity
but something that's
interesting is that quantum
mechanics on the other
hand is an idealization
that you obtain in the
limit as entropy goes to
zero because in in in kind

of one of these purely
computational models of
physics the quantum
mechanical state of a
system is described in
terms of its multi-way
structure it's described in
terms of you know when i
have a kind of a branching
program like this let me
find with these like this
then you know i can divide
it up into these into these
sort of into these
simultaneity surfaces and
if i associate each each
state of the program as
being like the analog of a
quantum eigenstate and the
kind of path weightings as
being an analogous to the
amplitudes associated with
these eigenstates i can
quickly build up a
description of this
multi-way system in terms
of the evolution of some
discrete analog of the
Schrodinger equation and
it turns out you get a
theory that is kind of
equivalent of the
mathematically isomorphic to
standard quantum
mechanics out of it so
quantum mechanics is sort
of you know inextricably
bound up with with with

the with the with the
phenomenon of the
multi-way system but if
you take the entropy to
infinity here then you're
effectively then the
sophistication of your
of your equivalence
function becomes
arbitrarily large which
means that you can
describe any essentially
any pair of states as
being equivalent and so
it turns out that the
that actually the the
quantum mechanical case
corresponds to the zero
entropy limit whereas the
kind of general
relativistic case
corresponds to the
infinite entropy limit
but they're kind of two
different two
fundamentally different
notions of entropy one
of which exists at the
single-way level one of
which exists at the
multi-way level and
again the question of how
these things interplay is
partly why we're you know
why we're investigating
this and it's clear that
that that question has
links to these quite

foundational questions
in fundamental physics
the ideal point mass and
the ideal distribution with
its center of gravity and
all this Dave question for
you from you yes um okay i i
was can you hear yep go
for it okay good um yeah i'd
like to to veer down to
the low road during this
discussion uh maybe show
business when people are
trying to explain what is
computational reducibility or
irreducibility often you'll
see a graph that says well
now here's the the
computation our target running
along the fox is running
along and behind it there's a
team of algorithms that
would like to catch it and
yeah it either does or it
doesn't outrun all of them
but some can come can seem
to come pretty close to
catching it uh now there's
something else people have
been looking at for a few
years the mandelbrot set the
mandelbrot set really is a
determined not only a
deterministic computation it's
a crisp computation every set
every point either is inside
the safe zone it's going to sit
there and it's happy and quiet

and it's you color it black
or it flies off to infinity
so you could it could just be a
bunch of yes or no's but
that's not the way people
trying to calm down after a day
of work want to look at it they
want to say hey i want to see
the 32 million deep mandelbrot set
and i want to see it in colors so
when you get the colors you ask
how hard did i have to work how
long did i have to grind along
before i made that decision oh
this is a point that's going to
settle down and be quiet and
uninteresting or it's going to fly
out to infinity so you put colors
it now i just wonder would it be
interesting to anyone to fuzzify
these gorgeous causal and
multi-causal graphs and just show
this is the portion where we had to
work really hard rule 35 or what we
had to go 15 000 generations before
we gave up and said we're not going
to follow this anymore the other one
oh after 30 steps i i see
that's a really interesting question
and and i mean yes so uh i mean a
couple comments on on on that i mean
so what one is i mean you're right in
the sense that yes the mandelbrot set is
a very clean thing to describe i would i
would say it's not i mean it's i would
argue it's actually not a crisp computation
in that sense for precisely for reasons
of computational irreducibility because

as you go arbitrarily close to the
boundary of the set you can have
complex numbers that stay that have a
kind of indefinite period of transient
right there's there's no upper bound on
how long the you know z n squared plus
c uh orbit can can last before it either
diverges or converges to to you know to
zero um and and and that statement that
you know that there are there will be
points that can you know that can remain
that can get tied up in these orbits
indefinitely is really a computational
irreducibility statement um but yeah i
mean your your question about uh you know
could you could you construct so i mean
that that that way of coloring points that
are on the boundary of the mandelbrot set
in in that way that this the so-called escape
time algorithm right where you kind of
where you color them based on how many steps
that i need to do before it either converged
or or escaped off you know above some or
you know the complex numbers modulus
uh exceeded some value um yeah it's an
interesting idea that you could try and
kind of construct geometrical representations
of the space of possible computations based
on a kind of escape time algorithm for
computational irreducibility yeah so i mean
there are the possible ways that you could
do that right that you you say you know we
know we've known since the days of turing
that the halting problem for turing machines
is undecidable right there's no finite
computation that you can do that will
determine with an arbitrary computer program
will terminate in finite time so you could

do it you could if you once you have a way
of kind of geometrizing the space of possible
computations which we which we have now as
you say um yeah you could easily construct
a kind of escape time algorithm where you
know you have your analog of the mandelbrot
set which is you know here's all the
halting computations here are all the
definitely non-halting computations and
then there's some boundary of very fuzzy
stuff where we kind of don't really know we
have to do a lot of work and even then it's
only heuristic um which is which is so
yeah i mean very directly analogous um and
then yeah and then the question becomes you
know so in the in the case of the m set you
know that there is some underlying theory
mostly i think mostly due to like duadi and
hubbard and people uh that allows you to kind
of predict like you know if you if you have a
finite filament you can like of the of the
mandelbrot set you can kind of predict you
know where that where that filament is going
to be based on some complicated complex analysis
argument and you know i mean the question
is once you if you once you have a
geometrization of the space of possible
computations can you construct some general
theory like the theory you know developed by
by duadi and hubbard that tells you things
about you know the topology of the computations
where the certain regions of the of the space
are connected whether they're compact whether
you can make predict you know whether you can
do a similar thing where you can make predictions
based on the geometry of one part of the set
you know where you can extrapolate to things
about the geometry of another part of the set

that's a really interesting question um and uh yeah
again it's as with so many of these things it's
it's one of these like i i don't know the answer
but it's a good reason for investigating you know
for pursuing the program right yeah from the
wolfram side as well as from the active
imprint side there's both the information
topology and the information geometry side
and not just in the kind of topological deep
learning but rather looking at the topology of
information flows which has been heavily
developed in the category theory related to
quantum information sciences and then the
information geometries that allow us to do like
machine learning type accelerated optimization
and gradients all these kinds of concepts that
come into play with geometry and we just don't get
them from topology so topology kind of sketches
the skeleton and then with the computers that we
have the information geometry is at least on the
data sets and the ways that we have computation
today that's kind of like the quantitative numerical
versus the formal um one question is are these all
discrete state space discrete time formalisms because in
active inference we often deal with hybrid models that
have discrete and continuous state spaces and the same
generative model or the same system of interest could be
modeled with like a discrete time chapter seven or a continuous
time chapter eight model so how does this deal with that
that's a really good question i mean yeah so so so most of what i've been uh
looking at so far has as consistent of um discrete time discrete space uh uh models
for no particularly principled reason other than um they're easier to analyze
right that that that for the most part because you can do explicit computations
because they're kind of more amenable to constructive analysis um it's easier to do
but the beauty of a lot of this you know that's one of the beauty of using beauties of using kind
of general mathematical formalism is that once you develop it um it's often quite easy to extend
even to into cases that you know to extrapolate to the cases that you didn't explicitly analyze so
um in principle this formalism works for for you know for for continuum space and

continuum time systems as well just with some you know with some slight modification so
rather than having say branching and merging you instead have you know if you think about
this thing as now being a dynamical system you know described on some symplectic manifold
then these kind of branching merging operations of the multi-way system become uh effectively
divergence and convergence operate you know differential operators are defined on the symplectic manifold
and uh so you know one place where we can start to analyze that explicitly and which i've done a
little bit of work on but it's one of these things which i want to go back to very soon um is looking
at petri nets so petri nets are interesting because they are a discrete time discrete space system but they
admit a continuum you know continuum space continuum time description in terms of ordinary differential
equations and so on so they're they're a nice example of a hybrid kind of discrete event versus
continuum event system where it's clear that this formalism can be used and and is somehow agnostic
as to whether the you know whether the underlying system is discrete or continuous um again there's
a broader philosophical point to make here which is that um in a way one of the reasons i don't
feel embarrassed to be working primarily with discrete systems is because you know again once you start to
think about things in terms of okay you have to not just care about nature you also have to care
about the computations the observer can perform and what it's able to infer um then you quickly realize
that in a sense just like whatever is you were saying earlier daniel you know uh beauty is in the
eye of the beholder i think discreteness and continue and and and continuity are also in the eye of the
beholder right i mean so if you have um if you have a you know a universe that is fundamentally
continuous that's described by you know a continuum lorenzian manifold or something but you're constrained so
that the only experiments you perform have computable outcomes have discrete outcomes uh where you know
whether with accountable with the possible number of um uh you know of observables is always countable
then in a sense it doesn't matter right it's it's irrelevant to you as an observer whether that we
know whether the whether the system is discrete or continuous uh because you know the only parts of
it that you can interface with and interact with are discrete and so you could have replaced
the underlying substrate with a purely discrete mathematical structure and you wouldn't be able to tell
so in some sense uh i i yes i don't feel too embarrassed discreet dealing with discrete event
systems because even if i don't necessarily believe that nature is discrete because i don't think that's
i'm not even sure how we would be able to answer that um i'm reasonably convinced that that you know the
the experiments that we can perform and the observations that we're able to perform are ultimately
computable and therefore you know the underlying substrate might as well be discrete even if it's not
sort of in reality so to speak yeah that's a great comment definitely calls back to your earlier
points about like discretization being in the eye of the beholder like in the active inference models
observations raw data may already be discretized depending on the situation but even if it weren't
like it were a continuous sensory perception or modeled as such analytically still commonly models
discretize and categorize as they move up cognitive hierarchies and that was like initially explored to

get more of this discrete either or decision making planning all those kinds of properties um well there's many interesting angles like i'm sure also it could be a multiplexed language model prompt but like what are you working on or excited about for 2024 uh that's a good question um so i've kind of already given some hints about you know the like this this general research program of of trying to understand uh computational complexity and algorithmic complexity and interplays between observers and systems through this category theoretic lens that's a yeah that's a that's a major thing which i started on say about a maybe a couple years ago i mean in some form i was i've been working on it for a long time but but this more recent perspective on it is maybe a couple years old but for various reasons over the last year or so i've kind of put that to rest and i've been focused on these uh much more physics oriented questions about discrete space time and understanding things like you know how the black holes work and how
does accretion work in discrete space now which is also very important and uh and and very exciting but uh i've sort of slightly been missing these more abstract directions and so i have maybe one or two major you know physics related things that i need to finish off and then i and then i really want to go back to this to to the greatest extent possible and yeah i mean what so one thing is that's quite clear is that there's uh um there's great interplay between this formalism and ex you know and existing theories of computational and algorithmic complexity so in particular you know so uh one very basic example is i mentioned before that you know you you have this kind of coherence between these two different algebraic structures between your the the operation of time-like composition versus the operation of kind of parallel composition and these two algebraic structures are in general related although the precise conditions that relate them are not clear and that's partly what we're trying to you know what we're trying to understand um but it turns out that that uh degenerate cases of that of that question corresponds to unsolved problems in computational complexity theory so for instance the p versus np problem can essentially be recast in these terms that you can recast the p versus np problem is the question about uh is the coherence between the time-like composition of computational complexity and the parallel composition of computational complexity which are what p and np respectively are really about it are those coherent conditions the the strictest they can be um which would be the case that that uh the p equals np or are they somehow more lax which would be the case that p does not equal np and so there's you know that that's that's one thing that we kind of have already investigated but it's clear that a whole bunch of questions about you know um how do time complexity and space complexity trade off or how do kolmogorov complexity and time complexity trade off these these are questions which can be recast in this kind of more algebraic category theoretic lens and and and um will hopefully give insight into this general program of trying to understand observers and their relationship to the world and those are kind of major um well with any luck those are major theorems that i hope we'll be able to prove at some point in 2024 um that's that uh that's that's awesome and it makes me think about parallel more nestmates more cpu threads deeper in times more sequential more planning and more cognitive single monolithic agent and then the kind of question is like can anything that a single agent mega matrix could do cannot be decomposable at space advantage or even at

space disadvantage in decomposed it into a single time step operation yeah that's a that's a super important question and one that you know uh with the possible exception of this community not many people have asked right i mean so uh that's something which comes up in quantum computing right so um a lot of the hype around quantum computation comes from these theoretical speed ups that derive from the fact that you're able to uh you know you're able to support these superpositions of different you know where you know each that each state of your data structure corresponds to a different eigenstate and you're able to evolve some super superposition of those eigenstates but then at the end you have to actually come to a definite conclusion about what the answer is you have to perform some measurement operation and that measurement operation is lossy it's often non-deterministic uh you you know you often have to repeat it multiple times um and you know it's it's becoming increasingly clear that for a large class of operations that were previously thought to have quantum advantage the additional complexity of the measurement step really kills any quantum advantage that you may have had that the you you know you get some advantage by doing unitary evolution uh but then you lose all of it by having to do the submission projection at the end and that's really a story of again this interplay between the time complexity saving of doing a multi doing a computation multi of doing a multi computation in parallel versus the loss that comes from the the complexity of the equivalence function that you need to apply in order to get to some definite uh conclusion about what the you know about about what happened because you know ultimately you need to somehow collapse that that that directed graph into a single thread uh of time in order to be able to have some coherent representation of what happened and so again understand that you know that's a place where understanding these you know these these trade-offs will become very important and as i say in some limiting case that that gives some perspective on your question daniel right which is which you know i agree is a very interesting question about you know um in principle we know that anything that a deterministic turing machine can do a non-deterministic turing machine can do and vice versa with some speed up or slow down um but that statement which is a classic result in you know in computability theory neglects all consideration of the equivalence function so there may be cases where the equivalence function is so complex that essentially the you know the the to do state equivalence becomes undecidable and so in that case you have a scenario where actually you you know you've got a multi-computational system but to to collapse it to one that's equivalent to a single-way system requires unbounded amounts of computational effort and so that so actually they become an equivalent even though you know computability theory says they should be the same um so uh it's clear that there's a there's a a a a a more rich more subtle theory that's underlying here that we're just beginning to kind of glimpse and that i hope we'll you know we'll be able to kind of to prove some new limited results about soon once we understand it a bit better awesome and i think definitely a special shout out to all of our colleagues on either like the wolfram and or active inference side because we've seen few if any active inference models phrased analytically or computationally with the wolfram technology um from studying complexity in other areas it's really clear to see how productive and powerful the software and the tools can be and changing and growing every day so it's really interesting maybe someone can um if they're listening this far in like go from one side to the other and back or make a wolfram active inference model or do some other kind of combination because it's very very fruitful territory and we know that our um elders have already spoken they've okayed it no but really it's so much it's so rich with connections here between um the areas that we're all studying and feeling like converging on many common places to to scaffold and jump off from together yeah i i i agree and um i mean at least the i mean the the kinds of things that we were discussing here about you know speculative execution and and behavior formation through free energy principle and so on those things should be relatively easy to implement in the framework that's already been developed here i mean so so you know that that's that's just a you know that's a question of just implementing some kind of computation of expected free energy and using that to you know to to to weight multi-way paths in the speculative execution model um so at least the beginnings of of that implementation i think the the path is pretty clear and we will probably end up doing at some point in the future anyway uh as part of as part of other research um so yeah so i i agree it's that's it's a very interesting it's a very exciting kind of um point of interface yeah well i hope that um we can stay in touch if you ever want to come back for a uh double oh 9.2 or if we want to even facilitate some kind of working group or some connections to really strengthen and like include the participation of more people in this super exciting area that would be amazing yeah that sounds fun let's let's try and set something up well dave first penultimate comments then jonathan you can have the kind of last comments yes um i hope you do get to continue on the wolfram physics side to think about a more general notion of what these ultimate things are uh are they observers or does that already prejudice the case of what you might find if you call them workers or actors or you know go back and and ask stewart kaufman what must a mind do to earn its way in the world let's make this keep happening thank you thank you dave for suggesting it also it was a great suggestion jonathan no and that's i think that's a fantastic note to end on i mean and and that that in a sense this this idea that we should start to move i mean so it's okay big picture for a moment like we're you know this formalism is being developed the the formalism i've described in this uh in this discussion is being developed you know assuming a kind of purely passive observer idealization and that's already been incredibly difficult right this it's clear there's a lot we don't understand of that but of course david is right that that in a sense you know what you know ultimately we want to start transitioning to a to a participatory observer model where where you allow for two-way interactions or you know higher order interactions between observers and systems and things that don't just go in one direction and yeah you know in a sense i view a lot of what we're trying to do with you know trying to nail down these notions of causality trying to understand these interplays between different complexity and entropy measures as you know the necessary groundwork for developing that subsequent theory right that does you know it's clear that if we want to have a a version of the you know of this kind of compositional multi-way formalism that is also compatible with things like second order cybernetics then uh you know at the very least we need to have a very coherent notion for what causality is and and you know and a robust algebraic description of that that's not going to to break or change and so i think um the the the non-participatory a participatory observer model is a useful starting point because it's one that is just within our grasp of uh you know of being being kind of kind of mathematically tractable um and then the hope is that the technology and the the ideas and the conceptual structure that we develop for understanding that will then as i say lay the groundwork for for for developing something that's that's more like what real you know active participatory observers do um and uh and yeah i mean i i think i i don't on the on the scientific side i'm not really sure i have anything any final comments apart from just you know as it's obvious this is a you know this is still a story that's being that's being developed and uh and yeah as daniel alluded to i hope that we can continue to uh to interact and and collaborate where where that makes sense and yeah at the very least i think incorporation of um kind of these yeah these active inference models within these uh discrete time in in the first instance discrete time uh you know computational frameworks uh and allowing things like speculative execution and multi-way path weighting based on um free energy estimates uh i think that's you know that that's a um that's a project that's of obvious mutual interest and and and something that i that i that i hope will happen in the coming months thank you we just speculatively executed active wolfram prince basically sounds excellent all right thank you jonathan thank you dave thank you everyone see y'all next time

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Thank you.